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Abstract: *The ever increasing sophistication of cyber threats has made traditional signature based Network Intrusion Detection Systems (NIDS) inadequate and ineffective. This review explores the shift towards Artificial Intelligence and Machine Learning in these NIDS, investigating how supervised learning models excel at detecting known attacks while unsupervised techniques identify unknown threats through anomaly detection. This review also shows superior detection rates of AI-driven approaches supported by figures from notable sources. However, critical analysis details the vulnerabilities of this approach through adversarial machine learning, where attackers craft malicious data to poison training sets or evade detection. The conclusion focuses on how AI presents powerful cybersecurity evolution, deployment must incorporate robust defensive strategies including adversarial training and explainable AI to ensure network security against evolving threats.*

Statement of AI usage: *For this project I utilized the power of AI to summarize information from papers and various sources to find the most relevant facts and figures to help reinforce my points. It was also utilized to help in the understanding of any unknown complex topics and definitions to give simpler explanation to improve my understanding of these definitions.*

1 Introduction

In the field of Cyber Security, organizations face new challenges from a wide range of attack vectors including Advanced Persistent threats and polymorphic malware (Liao et al., 2013). The transformation of critical digital infrastructure and the increase in Internet of Things devices have expanded the attack surface exponentially. Traditional signature based Network Intrusion Detection Systems (NIDS) operate on a reactive model, And relies heavily on pre-defined patterns and known attack signatures to spot malicious activity (Chiba et al., 2016). This approach proves insufficient against zero-day exploits and polymorphic malware that constantly alters its code signature to avoid detection.

The integration of Machine Learning (ML) and Deep Learning (DL) has completely revolutionized intrusion detection, creating a shift from reactive to proactive threat identification through behavioral analysis and pattern recognition (Muneer et al., 2024). Supervised learning algorithms such as: Random Forest, Support Vector Machines (SVM), and Gradient Boosting Machines have demonstrated exceptional performance, with detection accuracies exceeding 99% for specific attack categories on benchmark datasets such as NSL-KDD and CIC-IDS2017 (Lu et al., 2024). Simultaneously, unsupervised learning techniques have proven to be very efficient when identifying zero-day attacks and other threat methodologies that lack predefined signatures.(Mirsky et al., 2018)

However, this technological advancement introduces a critical vulnerability being adversarial machine learning. The data orientated nature of ML models makes them susceptible to carefully crafted attacks designed to bypass detection systems (Jmila and Khedher, 2022). This creates a massive problem within the Cyber security where defensive advancements leads to increasingly refined offensive techniques.

The widespread usage of AI in Cyber Security has created an urgent need to understand both its capabilities and limitations. This review aims to provide a balanced perspective on how AI technologies are changing network security while highlighting the important security challenges that come from the very tools meant to protect digital infrastructure.

2 Core AI Methodologies in NIDS

The application of Machine Learning in Network Intrusion Detection Systems is broadly categorized into two main approaches: supervised learning for detecting known threats and unsupervised learning for identifying previously unknown attacks (Thakkar and Lohiya, 2023).

2.1 Supervised Learning: Pattern Recognition for Known Threats

Supervised learning models operate on labeled datasets where each network traffic instance is pre-categorized into classes such as "normal," "DDoS," "Probe," or other specific attack types (Thakkar and Lohiya, 2023). Benchmark datasets including NSL-KDD and CIC-IDS2017 provide foundational training material. These datasets allow models to learn complex relationships between network features such as packet size, protocol type, flow duration, TCP flags, and payload characteristics and their corresponding security classifications.

Common algorithms deployed in supervised NIDS demonstrate varying strengths: Random Forest constructs multiple decision trees and achieves robust classification through ensemble voting; Support Vector Machines (SVM) identify optimal hyperplanes separating different attack categories in high-dimensional feature space and Gradient Boosting Machines sequentially build decision trees to correct errors from previous models (Al Lail et al., 2023).

Recent studies show exceptional performance, with F1-scores exceeding 99% for specific attack categories including denial-of-service attacks and network probing attacks. However comparative analyses shows a large variation in performance across different attack types. The primary strength of supervised learning approaches is their high precision and low false positive rates for attack types well represented in training data. Nevertheless, these models demonstrate a critical limitation they are unable to reliably detect zero-day attacks or novel threat variants that differ significantly from patterns learned during training (Buczak and Guven, 2016).

2.2 Unsupervised Learning: Discovering the Unknown

Unsupervised learning models work with unlabeled data, making them particularly useful for identifying new attacks without the knowledge of their signatures or patterns (Mirsky et al., 2018). These techniques focus on establishing the baseline of normal network behavior and then flagging any significant deviations from the baselines that may indicate malicious activity.

The key techniques include clustering algorithms such as K-Means and DBSCAN, which can detect coordinated attacks, unusual communication patterns, and behavioral anomalies. More advanced approaches use autoencoders, which are neural networks that learn to compress input data into its essential parts and then rebuild it as accurately as possible (Andresini et al., 2021). When anomalous traffic patterns are processed through with an autoencoder trained exclusively on normal traffic, the reconstruction error increases significantly and acts as an anomaly score.

Recent implementations demonstrate promising results Mirsky et al. (2018) developed Kitsune which is an online autoencoder-based system capable of detecting new attacks with 94% accuracy while keeping sufficiently low computational overhead for real time deployment. The foremost strength of unsupervised methods is their potential to detect zero-day attacks and novel threats lacking predefined signatures. However unsupervised methods have their own challenges with higher rates of false positives compared to supervised methods (Ali et al., 2024).

3 Hybrid Systems and Adversarial Vulnerabilities

3.1 Integration of Complementary Approaches

Modern systems often utilize hybrid approaches that leverage the strengths of both supervised and unsupervised methods (Muneer et al., 2024). In these multi-layered architectures, supervised learning models form the first line of defense, filtering network traffic to identify known attack patterns with high accuracy and minimal false positives. The remaining suspicious traffic is then subjected to unsupervised anomaly detection.

Studies show the effectiveness of this layered strategy, such as Tama and Lim (2021) implemented a hybrid system combining Random Forest for initial classification with DBSCAN for anomaly detection, achieving 97.8% overall detection accuracy while reducing false positive rates by 43% compared to using unsupervised methods.

3.2 The Adversarial Threat Landscape

Despite these improvements, the data dependent nature of Machine Learning powered NIDS introduces adversarial machine learning vulnerabilities (Papernot et al., 2016). The performance and reliability of these systems depend on training data integrity and the stable feature pattern representations.

Evasion attacks are the most common threat deployed on Machine learning NIDS. These attacks occur during the operational phase, where attackers subtly modify malicious network traffic

designed to cause misclassifications while preserving the attack's original malicious functionality (Goodfellow et al., 2015). Research by Papernot et al. (2016) demonstrated that these small changes could reduce detection rates from 95% to below 20% while maintaining attack effectiveness.

Poisoning attacks on the other hand, target the model during its training phase by injecting carefully crafted malicious samples into the training dataset (Biggio et al., 2013). This strategic corruption of training data causes the model to learn incorrect patterns or relationships, fundamentally compromising its integrity from the beginning. Recent studies indicate that poisoning as little as 3% of training data can reduce model accuracy by over 30%, demonstrating the fragility of ML systems to training data manipulation.

The accessibility of adversarial attack tools has lowered the barrier for executing these attacks. Open source options like CleverHans and Adversarial Robustness Toolbox offer pre-built implementations of attack algorithms, enabling even moderately skilled attackers to generate sophisticated adversarial examples against ML-based NIDS (Papernot et al., 2016).

4 Critical Analysis and Future Directions

4.1 Evaluating Current Limitations and Trade-offs

The current state of AI-driven NIDS reveals several limitations. First of all the reliance on benchmark datasets creates potential concerns about generalization. Ali et al. (2024) notes that models achieving 99% accuracy on standardized datasets often demonstrate significantly lower performance when deployed in production.

Second, the trade-off between model accuracy, adversarial robustness, and computational performance presents unavoidable challenges (Madry et al., 2019). Models designed to adversarial attacks typically sacrifice 5-15% detection accuracy compared to standard models while requiring substantially greater computational resources.

Third, the dynamic nature of network environments and continuous changes in attack techniques generate a requirement for frequent model retraining and validation, creating a significant workload for security teams. Furthermore, the quality and variety of training data significantly impact model performance. Biased or incomplete datasets can lead to models that perform well in laboratory settings but fail in production environments, creating a false security confidence (Ali et al., 2024).

4.2 Defensive Strategies and Research Directions

Current research focuses on several promising defensive approaches. Adversarial training involves intentionally incorporating known adversarial examples during model development to enhance resistance against evasion attacks (Goodfellow et al., 2015). This technique exposes the model to manipulations during training, enabling it to learn decision boundaries that remain stable under attack.

Explainable AI (XAI) represents another important defensive direction, developing human-interpretable machine learning models that provide transparent decision making processes (Muñoz González et al., 2017). Techniques such as LIME and SHAP enable security analysts to understand which network features most strongly influenced classification decisions, faster incident response and improving model refinement.

Ensemble diversity and feature randomization offer additional defense to systems. By deploying multiple models trained on different feature sets or using different algorithms, organizations can create systems where adversarial examples created to evade one model may still be detected by

other models.

Additionally, continuous monitoring and model updating are essential strategies for maintaining long term effective security. As new network traffic patterns and attack techniques emerge, static models slowly become less effective. Creating automated retraining pipelines that incorporate recent threat intelligence helps maintain detection accuracy over time while adapting to changing adversarial tactics (Jmila and Khedher, 2022).

Future research must prioritize hybrid defense approaches, by combining these strategies into comprehensive layered defense architectures. Particular attention should focus on developing lightweight adversarial defenses suitable for limited resource environments and creating standardized methods to measure robustness.

5 Conclusion

Machine learning has fundamentally changed Network Intrusion Detection Systems, shifting from reactive signature-based approaches to proactive behavioral analysis and anomaly detection. Supervised learning excels at detecting known attack patterns with remarkable accuracy, while unsupervised methods provide crucial protection against zero-day threats. The integration of these approaches in hybrid systems represents the current optimal practice in network security.

However, this technological advancement introduces significant vulnerabilities through adversarial machine learning, where sophisticated evasion and poisoning attacks exploit these models data dependent nature. The ongoing battle between detection and evasion techniques requires continuous innovation in defensive strategies, including adversarial training to improve model robustness and explainable AI to provide transparency and enable human oversight.

The human element remains the most important factor in AI-powered security systems. While automated detection provides scalability and speed, human expertise is essential for understanding complex threats, managing false positives, and making strategic security decisions. The most effective security operations will likely combine AI-driven automation with human analytical skills in a collaborative approach.

As cyber threats continue to grow in sophistication and scale, developing reliable, resilient, and adaptive AI-powered NIDS remains critical for protecting modern network infrastructures. Future research must focus on creating systems that not only detect known and unknown threats but also resist determined adversarial attacks while providing transparent, explainable decisions that security professionals can trust and act upon.

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